

1(a)	(red to magenta shows) increase in, pH / decrease in, concentration of, carbon dioxide ; (in the light <i>C. vulgaris</i>) uses up / AW, carbon dioxide for photosynthesis ;	2
1(b)(i)	light intensity ;	1
1(b)(ii)	any eight from: 1 use, lamp / AW, at minimum of 5 different distances from plant or use, lamp / AW, with minimum 5 different power ratings / wattages (at same distances) ; 2 use, a light meter / an app, to measure light intensity ; 3 <i>idea that</i> carry out investigation, in a darkened room / with no other light source ; 4 same / stated, mass / number / size, of alginate beads ; 5 same / stated, volume / concentration, of hydrogencarbonate indicator or same starting, colour / pH, hydrogencarbonate indicator ; 6 use a method to reduce heating effect of lamp ; 7 method to measure, colour / pH, of hydrogencarbonate indicator ; 8 measure / note / record, the colour/pH (of the indicator) after a, set / stated, time or measure / note / record, the time taken (for the indicator) to be a, set / stated, colour / pH ; 9 use at least three measurements for each light intensity and calculate a mean ; 10 safety: hazard and risk and mitigation ;	8

Question	Answer			Marks
1(b)(ii)	hazard	risk	mitigation	
	hydrogencarbonate indicator	irritant / allergy	gloves / goggles / mask / PPE / medication	
	alginate beads / <i>C. vulgaris</i>	irritant / allergy	gloves / goggles / mask / PPE / medication	
	heat (from bulb)	burns / injury	gloves / do not touch	
	11 use of a suitable control ;			
1(c)	will turn orange / yellow or pH decreases ; respiration, produces carbon dioxide ;			2
1(d)(i)	correct working ; 229 000 ;			2
1(d)(ii)	1 correct number of cells counted / 24 ; 2 correct working ; 3 correct calculation of number of cells per cm ³ / 240 000 ;			3

Question	Answer	Marks
1(d)(iii)	any two from: <i>Secchi stick is less accurate:</i> 1 subjective / difficult to determine when circle is no longer visible ; 2 error of ± 0.5 mm in the ruler ; 3 Secchi stick, not held vertical ; 4 no replicates / no repeats of investigation / no mean calculated ; 5 <i>idea that</i> Secchi stick used is, different / AW, from Secchi stick used for calibration curve ; 6 <i>idea that</i> inserting Secchi stick, mixes / disturbs / AW, suspension ;	2